The Impact of Generative AI on Teaching, Faculty, and Learners

A snapshot of the progress in my understanding so far
(January 2026 update)

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Abstract

Over the past two years, I have tried to conduct an in-depth analysis of the transformative impact of Generative Artificial Intelligence (GenAI) on higher education, highlighting a critical shift from standardized instruction toward new paradigms of personalized and "augmented" learning. By combining academic literature with my own field experiments carried out at Università Cattolica, the University of Turin, and ESCP, this work explores human-machine interaction dynamics and contrasts two collaboration models. The first is the "Centaur" model, characterized by a strategic division of labor in which the human remains in control while alternating with the AI. The second is the "Cyborg" model, which implies a deep and continuous symbiosis across every sub-task. These theoretical models find empirical support in the analysis of more than 570,000 student-AI conversations, which reveal differentiated usage patterns, from a passive "oracle" stance to a collaborative "co-creator" stance. The evidence suggests that more integrated and intentional modes of use lead to better learning outcomes. The document also examines the profound redefinition of roles. On one side, the instructor shifts from a dispenser of knowledge to an "augmented facilitator" and an "orchestra conductor" of the learning experience, taking on the critical responsibility of ethical supervision (human in the loop) to ensure content quality and mitigate risks such as bias and hallucinations. On the other side, students are required to develop new foundational meta-skills, captured by the AI Literacy framework, which include not only technical prompting literacy but, above all, critical evaluation and self-regulation. I raise a warning about the risk of an "inverted pyramid": a tendency for students to delegate higher-order cognitive skills (such as analysis and creation) to AI while neglecting the fundamentals, potentially undermining cognitive development. Methodologically, the paper proposes innovative teaching strategies such as AI-enhanced *Flipped Classroom* and a pedagogy of co-creation, in which students actively participate in designing virtual assistants and usage rules. A key concept emphasized throughout is the *Cognitive Mirror*: using AI not to provide answers, but to reflect and structure students' reasoning, making logical steps and implicit assumptions visible. Finally, the work argues for a radical redesign of assessment systems, shifting the focus from the final product to the validation of the thinking process. It concludes by underscoring the urgency of a systemic institutional response. To avoid exacerbating existing inequalities and

to compete in a rapidly evolving global landscape, universities must move beyond individual experimentation and develop a true "digital pedagogy" that integrates AI as a virtuous supplement, not as a substitute for human interaction.

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1 First introduction

A couple of years ago I decided to devote a significant part of my academic meta-"research" (I always use quotes out of respect for those who do research seriously, and I add "meta" to emphasize that, like it or not, I mostly do little more than rewrite, in the words of my own understanding, the things I read and find interesting) to the transformative impact of Generative Artificial Intelligence in higher education, with particular attention to paradigm shifts in teaching, the role of faculty, and students' learning processes. By synthesizing (almost blending) the available literature, theoretical models, and my first field experiences, this overview aims to clarify the dynamics of human-AI interaction in education, using the organizational approaches "Centaur" and "Cyborg" as the main discriminants that have accompanied my reading of the phenomenon for some time. I have tried to reflect on the pedagogical implications of integrating AI into teaching practices, on the evolution of the instructor's role, and on the new metacognitive competencies required of students. This is what I understand so far.

2 Second introduction

The emergence of Generative Artificial Intelligence, and its near-universal availability through tools such as ChatGPT-5.2, Claude, Gemini, Copilot, and similar systems, is, in my view, having a profound and lasting impact on the classic paradigms of university education. Over the course of this year, in my discussions with fellow faculty and with students, I have observed mixed reactions: on one side, enthusiasm for new educational opportunities; on the other, concern about potential misuse.

In the background of this debate, a real paradigm shift is unfolding in how we teach and learn, perhaps the first one this substantial in decades. The adoption of generative AI in higher education raises fundamental questions about the nature of learning in the digital age. If students can instantaneously access answers generated by AI systems, what meaning do we assign to "knowing" or "understanding" a topic? How does the instructor's role change when access to knowledge can, at least apparently, be disintermediated? What new competencies must faculty and students develop to be successful in a technologically augmented environment? Despite the growing relevance of these questions, institutional responses still appear fragmented. In my early field experiments, prevailing attitudes among faculty oscillate between two problematic extremes: pretending GenAI does not exist, and passively absorbing it by lowering the bar for teaching. In the first case, resistance to technological adoption is sometimes driven by ignorance of the phenomenon and sometimes by a cultural resistance to considering technology worthy of academic attention. In the second case, we see a reaction that reduces asynchronous activities and retreats (I would have said "barricades") into frontal lecturing, in the hope of limiting opportunities for students to "cheat" with AI support. Trying to replicate the approach I used in previous attempts to make sense of complex phenomena, I followed the same path in this exposition: I began by describing the impact of (Gen)AI on teaching and pedagogical methods, then I tried to understand how it transforms the instructor's role, and finally how it could be (and is) modifying students' learning processes. My goal was to provide a conceptual framework that goes beyond a limitation I often find in the literature, which cites large numbers of practical cases in (pseudo/para) administrative domains without trying to go deeper into the cognitive and educational aspects that are being redefined at the same time. This is only a first partial attempt, but we will have time to refine and extend it.

3 New educational paradigms and human–AI collaboration models

3.1 Moving beyond standardized instruction toward personalized learning

Let me start from the end: almost unconsciously, the (explicit or implicit) adoption of AI in educational contexts is pushing us from a concept of standardized instruction toward models that are more personalized and student-centered (customer-centered?). New GenAI-enabled technologies can act as intelligent tutors, adapting content and learning pace to individual needs, or as always-available assistants for clarifications and deeper exploration. In this context, people often speak of "augmented teaching" to refer to "learning environments" in which AI tools flank instructors and learners, widening and extending the boundaries of teaching and learning. This transformation appears particularly accelerated in institutions that have developed blended learning models, combining synchronous modalities led by human instructors with asynchronous modalities supported by LMS platforms (Learning Management Systems) enhanced by AI systems that personalize the educational experience by tailoring content distribution to individual performance, style, and learning pace. Personalization not only keeps students engaged, but should also create a more inclusive learning environment, where everyone can progress at their own pace without feeling they are "missing pieces" along the way.

4 Human–AI interaction models in academia: Centaur vs Cyborg

As already emphasized in previous "studies", and as can be observed in any organizational context undergoing AI adoption, two main interaction models emerge to describe how tasks can be distributed between human and artificial intelligence: the Centaur approach and the Cyborg approach.

The Centaur approach assumes a strategic division of labor: the human remains in control of key content and delegates specific parts to the AI, using it as a support tool to strengthen the outcome. It is a collaborative model in which instructor or student and AI take turns, each operating where they excel, much like the mythological creature that combines animal strength and human intellect into one team.

The Cyborg approach , by contrast, entails an integration so deep that it becomes almost inseparable: human and AI work in continuous symbiosis on every sub-task, with the student, the instructor, or even the AI itself relying on the other party's outputs, evaluating them critically, and improving them, including from independently produced content. This "fused human-machine" model maximizes the AI contribution at every stage, like a cyborg in which biological and technological components operate in unison. A concrete example of how these theoretical models manifest in teaching practice comes from the "sprint" approach adopted at the Tuck School of Business (Dartmouth College) and described by Anthony et al. (2024). [1] In this format, students tackle short, intensive projects where they alternate the use of AI tools (such as ChatGPT, DALL-E, and others) with their own creative and analytical contribution. Students found that, although AI generated many viable ideas, the most meaningful inspiration often came from unexpected connections made by humans. This empirically confirms what is hypothesized in the formulation of both organizational models (and described extensively by Dell'Acqua et al., 2025 [2]): AI increases the "creative surface area", but it is human judgment that identifies and develops the most promising insights. These theoretical paradigms find empirical support in data collected by Anthropic (publisher of the Claude chatbot) in a large-scale study of interactions between university students and AI. [13] The analysis of more than 570,000 academic conversations identified four interaction patterns. Two align with the conceptual reference models, while two correspond to less "intentional" uses:

1. **Direct problem solving:** The user seeks immediate answers or solutions with minimal interaction (an "oracle" use, or "centaur" if limited to a specific domain).

2. **Direct output creation:** The user requests the production of elaborate content with minimal interaction (an "oracle" use, or "centaur" if limited to a specific domain).

3. **Collaborative problem solving:** The user engages in an active dialogue with the AI to solve problems (an "advanced centaur" or "cyborg" use).

4. **Collaborative output creation:** The user actively collaborates with the AI to produce content (an "advanced centaur" or "cyborg" use).

It is worth noting that these four patterns appear in similar proportions (each between 23% and 29% of conversations). This suggests that students naturally adopt different interaction approaches depending on the task they face, and it also suggests that these approaches may represent evolutionary steps in usage maturity. In educational practice, a more evolved and intentional Cyborg approach is likely to be the most fruitful: AI is used to amplify human capabilities without replacing them. This hypothesis is consistent with my early teaching experiments, where groups that used AI as a "tutor" generally obtained better outcomes than those that used it as a pure "oracle". In particular, using a Cyborg approach to integrate, complete, or critique a previously produced artifact (for example, a "human" answer to a given question) produced noticeably better results.

4.1 Empirical validation of the theoretical models: analyzing the student voice

The Centaur and Cyborg theoretical models I described find significant empirical validation in recent UK higher-education research that analyzed more than 531,980 student comments using AI-based textual analysis techniques. [7] This study, one of the largest quantitative analyses ever conducted on student voice in educational contexts, offers valuable insights into how students actually perceive and use AI interaction in their learning journeys. The analysis identified four core dimensions through which the voice of the "customers" (students) manifests in the AI era:

- **Participation:** active involvement in educational processes;
- **Strategy:** intentional approaches to learning;
- **Difference:** recognition of individual diversity;
- **Right:** claims of educational agency.

What emerges with particular clarity is how these dimensions align with the behaviors that characterize the Cyborg approach: students do not merely "consume" AI-generated answers, but develop a critical capacity for dialogue and continuous negotiation with artificial systems. Particularly significant are findings related to sentiment diversity: the analysis revealed substantial differences by gender, ethnicity, and disability status in perceptions of educational AI. Female students, for example, show more negative sentiment when discussing themes related to "voice as difference" (likely reflecting concerns about identity in technological contexts), while showing greater positivity on themes of "voice as right". These data suggest that adopting Centaur or Cyborg models is not neutral with respect to student demographics, and that implementation likely requires a more nuanced, diversity-aware approach. The research empirically confirms what the theoretical formulation predicts: the most effective AI in education is not the AI that replaces the human, but the AI that amplifies students' ability to articulate, analyze, and defend their positions. The fact that more than half a million student comments exhibit this variety of approaches and sentiments suggests that the future of higher education will not be characterized by a single human–AI interaction model, but by a diversified ecosystem in which different students naturally adopt different strategies depending on context, discipline, and individual characteristics.

5 Emerging teaching methodologies

These new human–AI interaction dynamics are inevitably influencing teaching methodologies as well. To capture the full benefits of AI integration, we need to rethink instruction and adopt methodological tools such as the *flipped classroom*, a configuration in which students study foundational concepts outside the classroom with the help of an AI tutor, and then use class time to compare what they learned, engage in interactive and applied activities, and participate in instructor-guided discussion. In this way, technology supports active and personalized learning, while the classroom remains a space for human confrontation in which critical and creative thinking are prioritized. In my early experiments adopting an AI-enhanced *flipped classroom* model, despite some resistance, I experienced a clear benefit from moving much of the theoretical learning outside the classroom. This increased the share of in-class time devoted to practical and collaborative activities. During in-class sessions, instructors (potentially "augmented by AI") can guide active learning activities such as discussions, group projects, and complex problem solving, providing immediate feedback and adapting activities based on learners' responses. This approach turns the classroom into a dynamic, interactive space where students not only acquire knowledge but actively apply it, developing critical skills. A further step, which I consider particularly promising, is what we might call a **pedagogy of co-creation**. If up to now we have imagined students who use AI, here the focus shifts toward students who design with AI, and in part design the AI itself within the course context. Instead of merely consuming answers, students are involved in defining roles, rules, constraints, and quality criteria for the AI support they will use. It is a shift in posture: from users to co-architects of the learning environment. Incidentally, it is also one of the most effective ways to make clear that AI is not magic; it is a system to be governed.

Emerging literature suggests that these new methodologies can lead to deeper and more meaningful learning, provided AI is integrated strategically and intentionally within a solid, appropriately "adapted" pedagogical framework. As with many other digital-technology applications, we face the paradox of having to adapt the world to the needs of the technology by building "envelopes" in which it can operate successfully.

6 Transforming university teaching

6.1 Focusing on higher-order skills

The widespread use of generative AI mandates a radical rethinking of teaching and course design. One of the most significant contributions is the possibility to concentrate activities on higher-order skills such as analysis, synthesis, and critical thinking, delegating some basic or repetitive tasks to AI. In concrete terms, an instructor could assign a complex problem and require students to:

1. Try to solve it independently first;
2. Consult an AI model (appropriately "trained" on specific or technical content) to explore an alternative approach;
3. Critically compare the two solutions in a written reflection or in a group discussion.

Activities of this type transform AI into a learning enhancer rather than a tool for cheating. Students learn with AI and not from it, developing meta-skills in evaluating automatically generated contributions. Various authors (including Roberto [3]) propose different concrete strategies to modify teaching "assignments" in the AI era, maintaining the focus on higher-order skills. It is certainly appropriate to imagine 'multi-level' assignments that require students to make connections between different concepts, apply theoretical frameworks, and develop recommendations based on critical thinking. Furthermore, it is necessary to modernize writing assignments by requiring students to examine data, evaluate complex strategies, and link the case analysis to concepts from supplementary readings. And again, it is important to review, integrate, and modify questions on case studies to stimulate a deeper analysis rather than standard answers. Obviously, all this entails effort and one of the adoption obstacles seems to be precisely the low inclination and availability of faculty to face this path. A significant data point emerging from the Anthropic study is that students already delegate primarily high-level cognitive functions to AI, according to Bloom's taxonomy: mainly Creation activities (39.8%) and Analysis (30.2%), while lower-level cognitive functions such as Application (10.9%), Understanding (10.0%) and Memorization (1.8%) are less delegated. This raises important questions: if students entrust AI with precisely the higher-order skills they should be developing, how to ensure they are not compromising their own cognitive development? The metaphor of the "inverted pyramid" used in the report is eloquent: a structure that rests on the most complex skills instead of foundational ones risks collapsing. Early experiences conducted personally on the field also show the effectiveness of experimental approaches that intentionally limit AI responses. For example, instead of allowing AI to respond directly to a question, one can ask the tool to suggest creative ways through which students can independently find the answer. This strategy stimulates lateral thinking and keeps AI in the role of facilitator rather than substitute for the cognitive process.

6.2 New learning supports

AI offers innovative teaching supports that extend the learning process beyond the space-time limits of the traditional lesson. For example:

- Advanced chatbots trained on course material, capable of responding to student questions at any time;
- Systems for generating personalized examples based on the student's level of understanding;
- Virtual assistants that provide immediate feedback on exercises and assignments.

These tools, when well-orchestrated in the instructional design, can significantly enrich the learning experience, offering continuous and personalized support that would be impossible to provide under a traditional approach. A particularly innovative example of AI-powered teaching support is described by Gayeski [4] who illustrates how AI can be used to create interactive and personalized learning scenarios. The author describes a project in which she used ChatGPT to develop a fictitious but realistic business case that students had to analyze. Subsequently, she used tools like Visla to create introductory videos and Copilot Designer for visual elements. This multi-media approach allowed for the creation of an immersive experience that students found extremely engaging. Even more significantly, AI was also used to simulate a business crisis halfway through the project (a data breach), forcing students to adapt to unforeseen circumstances, just as would happen in a real professional context. This type of dynamic simulation would be difficult to achieve (or simply too expensive) without AI assistance. A particularly interesting evolution that I feel compelled to suggest is the creation of **GPTs** (bots "trainable" on the basis of specific knowledge) powered by course materials. These specialized assistants can take on different roles in the teaching ecosystem: from "study buddy" to "student" with peculiar behavioral characteristics who questions the person studying, thus stimulating understanding through the mechanism of reciprocal teaching. This approach has a very desirable side effect: it shifts the discourse from integrity intended as a "prohibition" to integrity intended as "design responsibility". If a student participates in defining rules, constraints, and criteria for an assistant, it is more difficult for them to experience AI as a shortcut and easier to experience it as a quality tool. Co-creation thus becomes a gym for maturity: learning to set limits, build controls, declare sources, design verifications. In other words, it is a way to teach cognitive governance, even before technological governance.

6.3 Rethinking assessment

A significant rethinking should also be planned regarding assessment methods. Since generative tools are capable of producing textual works, codes, or solutions to standard problems, it becomes less meaningful to evaluate the student only on the final product obtained in a context in which operational limitations and other debatable "positions" prevent having certainty that a "support" tool has not been used improperly. Some institutions are exploring new assessment strategies that prioritize the process and reasoning skills. A concrete proposal is to introduce more process-based evaluations: for example, requiring students to document the steps (including any prompts and AI responses) through which they arrived at a solution, to then evaluate the quality of their learning path in addition to the final answer. Operationally, this is a point in which what was previously defined as *cognitive mirror* is relevant, activated with simple but very disciplining tasks. For example: having the student produce a first independent answer, then asking the AI to extract the underlying argumentative chain, highlight the premises, and signal where the conclusion does not truly descend from the data. Alternatively, asking the AI to formulate three credible counter-arguments and then having the student refute them, forcing them to strengthen their own logical structure. Another useful technique is the "metacognitive replay": the student describes the steps followed to reach the answer and the AI rewrites them as a sequence of decisions, with checkpoints and possible typical errors. The value is not the final answer, but the increase in awareness regarding the mental path. Mollick and Mollick [5] propose an innovative assessment method based on role-play simulations created with AI, where students must demonstrate not only theoretical knowledge but also practical application capacity. In this approach, AI acts as a 'mentor' guiding the student through a simulated professional scenario (such as a negotiation or a business presentation), providing immediate and personalized feedback. What makes this assessment mode particularly effective is that AI can adapt the scenario in real-time based on the student's responses, creating a highly personalized learning experience. The authors also provide specific prompts that educators can use to create these scenarios, facilitating the implementation of this innovative evaluative approach that privileges the decision-making process over simple memorization of concepts or assumptions. In this way, even if AI contributed to parts of the work, the student is rewarded for how they guided and integrated that contribution; therefore, for the role of "orchestrator" of the produced content which, as we will see, becomes also one of the possible evolutionary scenarios on the other side of the desk. At the same time, the use of AI can be channeled in a formative manner: instead of prohibiting the tool for fear of plagiarism, it is incorporated into the teaching activity with clear criteria, promoting academic integrity and awareness. This approach requires a new "classroom contract" that establishes in a clear and transparent way the limits and boundaries of AI use. Personally, I have adopted (and adapted) Ethan Mollick's proposal of an explicit contract with the students that defines the acceptable use of AI, distinguishing between support roles (brainstorming, drafts) and prohibited uses (completing assignments entirely), requiring students to document their interactions with AI to promote transparency and academic honesty and also asking for constant feedback regarding the "emotional" impact in using these new techniques.

6.4 Balancing automation and the human touch

An essential element is to maintain and indeed strengthen the human touch in teaching. Every time we entrust an aspect of pedagogy to AI, we should seize the opportunity to add an activity of human interaction in the space we have generated. For example:

- If a generative model provided written feedback on a piece of work, the teacher could organize a following short individual meeting with the student to discuss that feedback. This approach naturally connects to the need to rethink the assessment. If AI makes the reasoning explicit, then it is possible to evaluate with more precision not only the final product, but the quality of the process: capacity to articulate premises, internal coherence, management of uncertainty, quality of checks, revision capacity. In this sense the *cognitive mirror* is also an elegant "anti-cheat": not because it prevents the use of AI, but because it makes it transparent and above all transforms it into a quality requirement. To obtain a good result, the student must show they know how to lead the mirror, not how to be subjected to it.

- If AI manages part of the explanation of contents, the time in presence can be dedicated to discussions, group works and deep-dives that only human interaction can effectively stimulate.

In summary, pedagogy boosted by AI must be designed by carefully balancing automation with the active and social activity of students, so that AI is a means and not the end of learning. This principle is continuously underlined in ongoing experiments, where it is reiterated that "the content is not the AI", but rather the disciplinary content which, through the tool, is reprocessed, decomposed and assimilated (hopefully better) by the students.

7 The new role of the instructor

7.1 From knowledge dispenser to *augmented* facilitator

It was inevitable that this transformation would not touch (challenge?) also the role of the teacher, who sees (sometimes despite themselves), their role transformed from dispenser of knowledge to augmented facilitator of learning. If at one time the teacher was the primary source of information for students, today contents and explanations are accessible instantly (even if sometimes not always in an impeccable way) via GenAI. This does not imply at all the marginalization of the teacher—on the contrary, it emphasizes its most exquisitely human and pedagogical aspects. Freed from part of the routine work (such as preparing questions for basic tests, correcting multiple-choice tests or providing standard feedback), teachers can focus on what really matters: inspiring students, stimulating critical thinking and guiding the development of articulated, lateral and deep skills. In other words, AI in class frees up time for the teacher, who can dedicate more to individual mentoring, Socratic interaction, motivation and emotional support of students—dimensions of educational work impossible to automate but sometimes (for the lacking attitude of the teachers themselves) avoided.

7.2 The instructor as an orchestra conductor

The teacher becomes an orchestra conductor of the learning experience: designing environments in which AI is exploited effectively as a didactic resource, while maintaining however the helm on the educational objectives and on the values to be transmitted. For example, a professor could decide to use an AI tutor to allow for personalized exercises to be carried out outside the classroom, but it will always be them who:

1. Select the materials with which the algorithm is trained;
2. Set the rules of use (when students must work alone and when they can consult AI);
3. Integrate the activities based on AI within the broader learning experience.

Co-creation can be made very concrete through co-designed didactic artifacts. Some examples: building as a group an assessment grid and using it to configure an assistant that returns feedback consistent with it; creating a "course tutor" constrained to respond only citing assigned materials, with a mandatory section "what I don't know"; designing an agent that does not give solutions, but that asks Socratic questions and proposes calibrated micro-exercises; building a standard group prompt that includes context, educational objective, quality criteria, and verification tests. The point is not that the agent be perfect, but that the process of designing it forces students to clarify what it means to learn well in that discipline. From personal experience I can guarantee that the introduction of AI into teaching practices is not at zero cost: it requires work of redesigning (and rethinking) the teaching modality, materials and classroom dynamics. This represents (unfortunately, sometimes shamefully considering the fact that we as a teaching body are a privileged social class in cultural production) an obstacle to adoption, but the first experiences suggest that the obtainable advantage can significantly outweigh the initial cost, especially when AI is used in a structured way, assigning it specific roles (such as Tutor, Oracle or Consultant) within the teaching activities.

7.3 New competencies required

The teacher has the task of training students for a productive and responsible use of AI tools. This means teaching:

1. How to formulate questions (prompts) in an effective way;
2. How to verify the answers obtained;
3. How to recognize the limits of AI.

It is known to everyone by now that an accurate formulation of prompts can influence drastically the quality of the proposed solutions (and this remains true even if the quantity of traditional software layers that now drive questions toward LLMs in a 'guided' way is impressive), given that different strategies emerge even with slight variations in the request. The professor themselves, therefore, must acquire a new digital literacy—knowing how to dialogue with AI—and become a mediator of this competence toward students. Empirical observations conducted during this first "augmented" academic year highlight that many students, while quickly adopting AI tools, use them with direct prompts lacking context, description of the task, or operational instructions. This "mis-use" reflects a common dynamic in the adoption of digital technologies: the adoption that anticipates understanding and the proper "use". Unlike the traditional approach to learning new tools (study of the manual, identification of use cases, design of flows and only finally implementation), the adoption of contemporary technologies often inverts this process. Specific training on prompting, not in generic terms but aimed at the learning objective, becomes then an essential component of the new teacher's role.

7.4 Ethical and qualitative supervision

Another crucial aspect of the new teacher's role is to ensure the ethics and reliability of the training process in the AI era. The teacher acts as a 'human in the loop', or rather remains the ultimate supervisor who verifies and approves the contributions of AI before they impact the training path. AI systems in fact can (sic!) incur in hallucinations (which, simplifying, we could define as invented but plausible answers) or present bias not immediately evident; without an adequate human supervision, there is the risk of introducing misinformation or distorted messages into the learning process. For this reason the teacher, more than in the past, becomes the guarantor of the quality of knowledge with which the student comes into contact. They have also the responsibility to discuss openly with the class the ethical implications of using AI, modeling a critical and conscious attitude.

7.5 AI as a supplement, not as a substitute

Studies in the field of EdTech suggest viewing AI as a supplement, not substitute for human teaching. Even the best virtual tutors tend to function in an optimal way flanked by the guidance of a flesh-and-blood instructor, and not left to operate in total autonomy. The centaur ideal (human+AI) in the teaching sphere means exploiting the potential of automation and of data analysis without losing the empathy, creativity and judgment that only a human educator can bring. The value of the teacher remains therefore central: AI can assume secondary tasks, but the inspiration, the guide and the example come still from the human being. Despite technological progress, the human element remains irreplaceable in education. Educators must balance the efficiency of AI with the need for personal interactions, empathy and individualized support. AI can enrich the educational experience, but it cannot replace the mentoring, encouragement and emotional support that teachers offer. My observations on the two predominant problematic attitudes in the teaching body ('pretending GenAI does not exist' vs. 'succumbing to it by adapting downwards') find a more systematic explanation in recent research that applied the Technology Acceptance Model (TAM) to the adoption of generative AI by educators [9]. What emerges from these studies is that the attitude of teachers toward AI is not simply a matter of technical competence or resistance to change, but reflects deep concerns about the impact on one's own professionalism and identity. The teachers I observed 'barricading' themselves on frontal instruction often express, when prompted in private conversations, the fear that AI could threaten what they perceive as their professional added value. It is a dynamic reaction: if a student can obtain personalized explanations from any ChatGPT at any hour of the day or night, what role remains for the university teacher? Research shows that this fear, while understandable, is often based on a limited understanding of the collaborative possibilities offered by AI. Teachers who develop a more positive attitude are those who manage to see AI not as a competitor but as a tool that can free up time for high-value activities: personalized mentoring, facilitation of complex discussions, design of innovative learning experiences. Paradoxically, AI can make the teacher's role more 'human', not less. However, this change of perspective does not happen spontaneously. Institutions that have had success in AI integration have implemented structured institutional work cycles that include:

1. Addressing immediate needs for policy and technical support;
2. Creating principled frameworks to guide the pedagogical rethinking;
3. Stimulating reflection on the broader academic implications of AI [10].

What I find particularly illuminating in these systematic approaches is the emphasis on 'multiple cycles of engagement': the change does not occur through a single training or decision, but requires an iterative process of experimentation, reflection and adaptation. Teachers need time and safe space to explore, make mistakes and learn. The most effective institutions have created pedagogical 'sandboxes' where teachers can experiment with AI without evaluative pressures, gradually developing competence and confidence. A crucial element emerging from these experiences is the importance of peer learning among teachers. The most significant changes occur when teachers see their colleagues using AI successfully, not when they are instructed from the top on how they should behave. This suggests that the most effective strategies for promoting AI integration could be bottom-up ones, based on the sharing of concrete experiences and results measurable rather than on theoretical proclamations or institutional obligations.

8 Student learning in the era of generative AI

8.1 The virtual study companion

On the student side, the integration of generative AI redraws the learning experience and the competencies expected from the graduates. Students of today find themselves with a virtual study companion always available: they can ask for clarifications, translations, examples or even an alternative explanation of a difficult concept, all through a prompt. This augmented learning offers great advantages, in particular for personalization: a student can proceed at their own pace, receiving additional explanations when remaining stuck and more complex challenges when showing mastery—all through an AI assistant that adapts to their level of competence on the specific subject. In many cases, AI can act as a 'leveler' (in homage to Totò's thought), helping those who have greater difficulty to bridge gaps of preparation (a bit like what emerged in studies on Knowledge Workers, in which AI tools improved above all the performance of the less expert participants). In the academic sphere this could translate into a reduction of certain initial gaps: every student has an additional personal tutor, capable of providing targeted support when the teacher is not available.

8.2 Meta-competencies, critical evaluation and prompting literacy

This apparent democratization of educational help brings with it new responsibilities for the student. It becomes fundamental to develop a meta-competence: the capacity to evaluate critically and verify the information provided by AI. Relying blindly on a generated answer can be dangerous, because the model could provide incorrect or partial content. In a teaching exercise in which students worked with an AI team partner, it was seen how acquiring experience in the critical evaluation of AI contributions constitutes a key competence for the future. Students are therefore required to learn to:

- Fact-check suggested solutions;
- Compare AI ideas with other sources and with their own knowledge;
- Discern when to follow the machine's advice and when instead to trust their own judgment.

In other terms, knowing how to ask good questions and knowing how to direct and validate the answers are essential skills in the era of Generative AI. A further emerging ability is 'prompt literacy': students must learn to communicate effectively with AI systems, articulating requests in a precise way to obtain useful results: this additional baggage constitutes a completely new form of digital literacy today indispensable. Making the context explicit, specifying roles ("explain how you would to a beginner..."), and iteratively refining questions are strategies that students must master to fully exploit a generative assistant. Not by chance, it is expected that the graduates of the next future will have to possess not only disciplinary competencies, but also the capacity to use AI as a collaborator in their own work processes. This will be true in the professional world—where already knowing how to cooperate with systems of AI increases employability—but must be cultivated already at the university, integrating it among the formative outcomes and activating in a conscious and explicit way a transformation of the attitude towards the continuous learning modalities. Getting used to rethinking the problem to solve in terms of sub-activities and contextually improving one's prompt ability through techniques like "chain-of-thought prompting" allows to exploit the available tools to the maximum, bringing the mode of execution of the algorithm closer to human cognitive processes and effectively increasing the absolute quality of the product (in this case with the double meaning of result and multiplication of efforts). In the last few months a more precise name has started to be used for a function of AI that we had already glimpsed empirically, but which tends to escape when we speak only of "tutor" or "oracle". It is called *cognitive mirror* the use of an AI not as an assistant that answers, but as a tool that reflects. In practice, AI is used to return to the student a reformulated, structured and interrogatable version of their own reasoning, making visible implicit steps, logical leaps, contradictions, undeclared assumptions. It is a subtle but decisive passage: AI does not substitute thought, it makes it observable (with a necessarily different point of view) making it effectively also improvable.

8.3 A structured framework for AI literacy

What I have observed empirically in my experimentations finds confirmation in a more structured theoretical framework developed recently for the integration of AI literacy into higher education [8]. This model, which we could define as "holistic", identifies four fundamental pillars that every student should master: **Knowledge** (understanding how AI systems work), **Use** (practical capacities for effective use), **Evaluation** (critical thinking about results), and **Ethics** (awareness of moral and social implications). What I find particularly interesting about this approach is that it goes beyond mere traditional "digital literacy" to embrace what the authors define as "meta-skills": competencies that allow for continuous learning and adaptation to evolving technologies rather than simply learning to use specific tools. It is a bit like the difference between teaching to fish and giving a fish (of evangelical memory): instead of training students in the use of a specific chatbot or a specific "generation" of AI, we prepare them to collaborate effectively with whatever AI system emerges in the coming years. The framework underlines the importance of guided and reflective self-learning: it is not enough to know how to formulate effective prompts, it is necessary to develop the capacity to reflect meta-cognitively on one's interaction process with AI. This means asking not only "what am I asking AI?", but also "why am I asking it in this way?", "what are my implicit bias in this formulation?" and "how can I improve my strategy of collaboration?". A crucial component emerging from this structured approach is the necessity of disciplinary contextualization: AI literacy cannot be taught in the abstract, but must be developed within specific contexts. A medical student will have to master the use of AI in ways very different from a student of literature or engineering. This observation strengthens what has already been experimented in my teaching activities: the effectiveness of AI integration depends crucially on the capacity to adapt tools and methodologies to disciplinary specificities. What strikes me most in this framework is the explicit recognition that universities are currently "lagging behind" in the systematic integration of AI literacy into their curricula. This delay is not only technological, but also cultural and pedagogical: it requires a deep rethinking of how we conceive learning, assessment, and even the role of the university institution in the era of AI. The good news is that this delay can transform into an opportunity: we still have time to do it well, avoiding the errors of a precipitous and superficial adoption.

8.4 Redefining the meaning of "learning" and setting reasonable limits

The theme that seems to me to still remain unresolved, however, is the fundamental pedagogical question: what does it mean to "learn" in an era in which GenAI can provide (illusorily) immediately solutions and contents? If on one hand it is useless to insist on tasks of pure memorization or mechanical calculation that a machine performs in an instant, on the other hand one does not want the student to become a passive executor of instructions from AI because in any case there is and continues to be value in the effort of the experience. The key lies in active learning: using AI to free students from some burdens, but committing them immediately after in activities of application, project, debate that require deep and lateral understanding regarding the contents to be learned. For example, a computer science student could use a generative model to quickly write the code for a standard algorithm, but the real task could consist in explaining why that code works, optimizing it or adapting it to an unprecedented case. In this way AI covers the basic tools of the trade, while the student trains on the more advanced reasoning and "engineering" creativity capacities. What emerges from the cases that could be verified empirically is that intensive use of AI in study requires a lot of maturity and self-management. With a virtual assistant always available, the student must learn to self-regulate: understanding when it is appropriate to ask AI for help and when instead to try solving in autonomy to strengthen their own abilities. Ladd [6] proposes five specific exercises to help students develop a balanced relationship with AI, avoiding both rejection due to fear and excessive dependence. Particularly relevant is the exercise 'say the same thing', which demonstrates to students how LLMs work, demystifying AI and helping them recognize that they can master it or at least direct it. Another significant exercise involves critical analysis of feedback generated by AI: students compare their own evaluation of a work with that produced by AI, così recognizing both the points of strength and the limits of artificial intelligence. Ladd also underlines the importance of preserving human competencies through activities that AI cannot replicate effectively, such as negotiation, debates, and collaborative problem-solving challenges. These practical exercises represent concrete tools to help students develop that capacity for self-regulation that will be fundamental in their academic and professional future. Awareness is needed to avoid shortcuts leading to a superficial learning. For example, if a student limits themselves to copying the solution suggested by AI without understanding it, they are depriving themselves of a formative experience; at the same time, ignoring or consciamente avoiding altogether the use of such powerful tools could mean losing opportunities for in-depth exploration. Pedagogy will therefore have to encourage a balanced use: AI as a brilliant "desk mate", but to be consulted with criterion and critical spirit, within a path in which the student remains active protagonist of their own training.

8.5 Virtuous uses of AI by students

Up to this moment I have observed few cases of truly "virtuous" use of AI by students of my courses, but these few cases show notable potential. The most effective uses can be categorized as follows:

1. **Synthesis and organization:** Students use AI to summarize texts, extract key information, and create indices, facilitating organization and navigation of study materials. AI can also categorize information, making study management more effective.

2. **Conceptual understanding and continuous comparison:** AI helps to visualize complex ideas creating comparative tables between different concepts, explaining flowcharts and graphs, and simplifying complex visual data for better understanding.

3. **Support for problem solving:** AI tools assist in problem solving, providing step-by-step tutorials and correcting solutions. They can also break down exam questions and answers into simpler parts, making difficult concepts more accessible.

4. **Preparation for exams and continuous improvement:** Students use AI to prepare for exams by practicing with simulated questions, receiving immediate feedback, and progressively improving their preparation.

5. **Resource searching and learning of new topics:** AI can retrieve specific information from study materials and suggest additional references, which is particularly effective with the "deep research" features of the most recent models. When tackling new topics, it can provide a list of fundamental terms and explain them in detail, adapting to the level of knowledge of the student.

At the same time, data collected by Anthropic reveal interesting patterns of AI adoption strongly differentiated among students of different disciplines. STEM students, in particular those of Computer Science, result to be early adopters, with a disproportionately high representation in interactions with AI (computer science students represent 36.8% of academic conversations despite constituting only 5.4% of graduates). On the contrary, students of Business&Management,

Medicine, and Humanities (including Law) show lower adoption rates compared to their representation in the student population. Significant are also the differences in the ways in which students of different disciplines interact with AI:

- Students of Natural Sciences and Mathematics tend to use AI mainly for problem solving;
- Students of Computer Science and Engineering show a preference for collaborative interactions;
- Students of Humanities, Business&Management, and Medicine present a more balanced use between direct and collaborative approaches but unbalanced on a direct (one-directional) use as an "oracle".

These differences suggest that the integration of AI into teaching should be calibrated based on disciplinary specificities, recognizing that different fields of study can benefit from differentiated human-machine interaction modalities.

9 Challenges for a conscious integration of AI in university

9.1 Academic integrity and assessment

The Anthropic study highlights that about 47% of conversations between students and AI are of the "direct" type, meaning they seek answers or content with minimal involvement. Although many of these interactions may have legitimate learning finalities, problematic conversations have also been identified where AI was used as a shortcut for answers to language or multiple-choice tests or for rewriting texts and essays to avoid plagiarism detection. Surely it is necessary to value the process and reflection more than the final result, for example, by requiring accompanying notes in which the student explains how they used (or deliberately did not use) AI in a task because the objective is not to demonize the use of AI, but rather to ensure that the grade continues to certify actual analytical, critical, and communicative capacities of the student. The adoption of a clear "classroom contract" represents a first concrete step in this direction, establishing explicit limits and boundaries for the use of AI sia in the classroom che nelle attività asincrone but alone it is not enough. Nell'attesa che l'evoluzione tecnologica incessante produca sistemi sempre più trasparenti (in grado di spiegare su quali fonti o ragionamenti hanno prodotto una certa risposta) e allineati ai valori accademici l'onere della validazione anche etica dei risultati ricade sulla comunità educativa che deve farsi carico anche di assicurare la conformità alle normative specifiche sull'AI e più in generale sulla protezione dei dati (come AIAct e GDPR in Europa). Are our universities ready to face queste ulteriori sfide e responsabilità o la cronica mancanza di qualcosa (fondi e competenze) spingerà a giocare al ribasso? Una delle dinamiche che andranno monitorate molto da vicino (per evitare che produca effetti negativi così come per evitare che possa diventare una facile scusa per un "luddismo di ritorno") è il fatto che l'introduzione massiccia dell'IA può accentuare disuguaglianze esistenti o crearne di nuove. Non tutti gli studenti hanno accesso allo stesso livello di strumenti (basta pensare a versioni avanzate di servizi AI a pagamento) o alle competenze digitali per usarli efficacemente. A concrete risk is that students già avvantaggiati (per background socio-economico o contesto scolastico precedente) traggano maggior profitto dall'IA, mentre altri rimangano indietro. Universities must commit affinché l'adozione dell'IA, quando prevista, sia inclusiva (anche se questo termine sta diventando poco di moda): fornire accesso equo agli strumenti, magari integrandoli in piattaforme istituzionali, e offrire formazione di base a tutti gli iscritti sulle abilità di utilizzo responsabile. Analogously, anche tra i docenti esiste un divario culturale e formativo nell'approcciarsi a queste tecnologie. Investing in continuous training programs on AI for academic personnel is essential per evitare che solo una minoranza di "tech-savvy" tragga vantaggio (e faccia trarre vantaggio agli studenti) dalla generative AI. La comunità accademica deve essere coinvolta nel definire come e perché usare l'IA, condividendo buone pratiche e risultati sperimentali positivi. Solo con una comprensione diffusa dei benefici (e dei limiti) si potrà passare da una fase di diffidenza a una di adozione consapevole. I believe che le teorie costruttiviste dell'apprendimento possano e debbano essere aggiornate per includere l'IA come elemento dell'ecosistema di apprendimento: ciò che spesso manca nel contesto accademico od in generale dell'istruzione superiore è un discorso aperto, critico ed innovativo sul come giocare al rialzo GRAZIE a queste tecnologie, superando gli atteggiamenti di negazione o di adattamento al ribasso attualmente predominanti.

10 Annotations of a field experimentation

10.1 Methodological and merit note on the experimentation

Come integrazione empirica del mio percorso di ricerca, negli ultimi due anni accademici ho avuto modo di sperimentare nei miei corsi diverse forme e sostanze di integrazione dell'intelligenza artificiale generativa nel processo di apprendimento. This article mira a riassumere gli elementi salienti delle sperimentazioni ed a riportare i feedback raccolti dagli studenti, con l'obiettivo di offrire spunti utili ad altri docenti interessati a intraprendere questa (non facile, ma gratificante) esperienza. L'esperienza si è evoluta attraverso diverse sperimentazioni, condotte nel mio corso di Management Control Systems and Performance Management presso l'Università Cattolica di Milano, nel corso di Management Information System presso l'Università di Torino e nel corso di AI & Data Transformation nel Master internazionale F&B Management di ESCP (campus di Torino).

La sperimentazione è stata condotta inizialmente utilizzando ChatGPT come strumento di supporto, poiché era quello con cui gli studenti avevano maggiore familiarità. Successivamente, volendo elevare il livello della sperimentazione, ho predisposto due GPTs (i bot "addestrabili" sulla base di una conoscenza "specifica") che ho addestrato con tutti i materiali del corso (le mie presentazioni, i miei articoli e gli appunti che a mano a mano gli studenti mi condividevano) e che ho "specializzato" tramite un apposito impianto di prompt per giocare due ruoli: quello del compagno di studi e quello dello studente. Tale sperimentazione è stata implementata sistematicamente in tutte le lezioni dei corsi. Ho costantemente sottolineato che il contenuto di interesse NON era l'AI in sé, bensì i temi del corso, e che l'AI doveva essere intesa come uno strumento, non come l'obiettivo. A partire dalle prime settimane di corso (ma non immediatamente), ho fornito alcune nozioni di base sul funzionamento degli LLM e sull'intelligenza artificiale (a livello macro, dato che il focus del corso era differente) e su elementi di prompt engineering finalizzati all'uso degli strumenti per gli obiettivi di apprendimento. Nel corso dei due anni di sperimentazione non ho apprezzato un "radicale" miglioramento da parte degli studenti circa le competenze di prompting: sicuramente una maggiore famigliarità con gli strumenti, ma un utilizzo ancora largamente "intuitivo".

10.2 Execution of the experimentation

La sperimentazione si è avvalsa in alcuni casi di un approccio di *flipped classroom* assegnando agli studenti materiali di studio (capitoli del testo o articoli) inerenti agli argomenti della lezione, da leggere e "studiare" in anticipo rispetto alla classe. In questi casi la lezione in aula era suddivisa in tre momenti:

1. **Initial moment of deep-dive and recap on contents studied independently.** Durante questo slot ho riassunto i messaggi chiave e inquadrato gli elementi fondamentali in un contesto logico più ampio (una struttura di riferimento adottata per l'intero corso).

2. **First moment of interaction:** che prevedeva la suddivisione della classe in tre gruppi con modalità di utilizzo diverse dell'AI per rispondere a domande sull'argomento della lezione.

3. **Second moment of interaction:** dedicato alla soluzione di un business case sull'argomento della lezione, svolto da due gruppi anche in questo caso con l'uso (opzionale) dell'AI.

In altri casi dove la frontale era comunque necessaria (assenza di un testo di riferimento, importanza della spiegazione dei concetti rispetto ai concetti stessi) la struttura tripartita è stata mantenuta, semplicemente con una quota tempo maggiore dedicata al primo momento. I tre gruppi, nel primo momento di interazione, sono stati strutturati secondo le seguenti modalità d'uso dell'AI:

1. **NO-AI:** Il primo gruppo rispondeva alle domande basandosi esclusivamente sulle proprie conoscenze, senza supporto del testo o dell'AI. Questo metodo ha permesso di valutare la capacità critica e la profondità delle informazioni acquisite nello studio individuale e nel recap.

2. **AI come Oracolo:** Questo gruppo formulava la propria risposta utilizzando esclusivamente le risposte generate da ChatGPT, con la limitazione di non poter modificare il contenuto generato ma solo valutarlo e criticarlo. Questa impostazione mirava a valutare la capacità del gruppo di formulare prompt chiari e completi e, al contempo, a stimolare la revisione critica delle risposte fornite dall'IA.

3. **AI come Tutor:** Questo gruppo seguiva una metodologia in due fasi: prima formulava una risposta autonoma, poi la forniva all'AI per ottenere feedback e suggerimenti di

miglioramento. Questa metodologia ha consentito agli studenti di rivedere e perfezionare le proprie risposte combinando ragionamento critico e utilizzo dell'AI.

I gruppi hanno ruotato, sperimentando ciascuno le diverse metodologie e rispondendo a ulteriori domande di riepilogo. Questo approccio ha permesso agli studenti di confrontarsi con vari modi di utilizzare l'IA, evidenziandone vantaggi, limiti e contesti di utilizzo ottimale. Nel secondo momento di interazione, i gruppi hanno risolto un caso studio utilizzando l'AI come "consulente." In pratica, l'AI era chiamata a NON rispondere direttamente alle domande del caso (compito affidato al gruppo), ma a suggerire cinque strategie alternative per affrontare la situazione aziendale descritta. Questa impostazione, che ho definito "AI come consulente," ha reso evidente l'importanza di una formulazione accurata dei prompt, poiché strategie di risoluzione diverse emergevano anche con lievi variazioni del prompt. I metodi precedentemente descritti sono stati ulteriormente sviluppati utilizzando GPTs condivisi con gli studenti e "addestrati" sui materiali del corso. Le risposte erano pertanto molto più specifiche e pertinenti ed anche il contributo nello studio è stato maggiormente rilevante. La parte più interessante della sperimentazione è stata quella condotta usando un GPTs addestrato sui materiali del corso ma "istruito" per comportarsi come uno studente, con delle caratteristiche "comportamentali" peculiari. Essendo infatti assodato che uno dei migliori strumenti didattici a disposizione degli studenti è "impersonare" il ruolo del docente di fronte ad uno studente "ingordo" di dettagli e nozioni, ho voluto sperimentare questo modello operativo addestrando un bot per fare domande (in modo interessato o "svogliato e ipercritico" a seconda del tipo di studente selezionato all'inizio dell'interazione) al fine di "farsi spiegare" un argomento. Alla fine dell'interazione (dopo una decina di botta e risposta) il chatbot da una valutazione su quanto quella "lezione" sia stata, dal punto di vista dello studente, efficace ed esaustiva. Nella scelta dei "ruoli" e dei meccanismi dialogici da stabilire mi sono evidentemente basato sulle risultanze di un mio precedente "studio" incentrato sulle tipologie di interazione uomo-macchina descritte (mi riferisco ai profili del Centauro e del Cyborg, ovviamente). La suddivisione dei gruppi in modalità di utilizzo diverse dell'IA (NO-AI, AI come Oracolo, e AI come Tutor) può essere vista come un tentativo di esplorare e replicare queste dinamiche comportamentali in un contesto didattico.

Centaur Approach: La modalità "AI come Tutor" si avvicina al comportamento del Centauro, caratterizzato da una divisione strategica del lavoro tra umano e IA. In questa modalità, gli studenti creavano risposte autonome e poi le rivedevano con l'AI, che agiva da strumento di supporto, suggerendo migliorie e approfondimenti. Come nel profilo del Centauro, il lavoro umano e l'IA si alternano, con gli studenti che conservano il controllo dei contenuti principali, utilizzando l'IA per affinare le loro risposte e per migliorare la presentazione e la struttura, specialmente in fasi iniziali e conclusive dell'analisi.

Cyborg Approach: Al contrario, la modalità "AI come Oracolo" riflette in parte l'approccio Cyborg, dove l'integrazione tra umano e IA è quasi inscindibile. Gli studenti formulavano prompt per l'IA e si basavano totalmente sulle risposte generate, limitandosi a criticarle e valutarle senza modificarle. Questo rappresenta una forma di collaborazione stretta e integrata, dove l'output finale deriva da un'interazione continua e indissolubile tra le risorse cognitive dell'IA e l'utente umano, simile al modello Cyborg in cui le due entità lavorano insieme su ogni sotto-compito.

10.3 Summary of student feedback on the educational experiment

Learning Prompt Engineering: gli studenti hanno apprezzato l'approccio metodico nella formulazione dei prompt. Ad esempio, uno ha evidenziato come, applicando la struttura CTI ("Context, Task, Instruction"), sia riuscito a migliorare sensibilmente la qualità delle risposte di ChatGPT, adottando velocemente questa tecnica: "Oggi, quando scrivo un prompt su ChatGPT, mi viene naturale seguire i tre punti chiave 'Context, Task e Information'."

Development of Critical Thinking: L'equilibrio tra teoria e pratica ha stimolato il pensiero critico degli studenti rispetto alle risposte di ChatGPT. Una studentessa ha apprezzato la metodologia, affermando: "L'adozione di questo metodo ha favorito lo sviluppo di un pensiero critico, incoraggiandoci a riflettere in modo autonomo e ad approfondire i concetti." Un'altra ha sottolineato l'importanza di utilizzare l'AI in modo consapevole, rilevando un aumento di coinvolgimento rispetto ai metodi tradizionali: "Permettere l'uso di ChatGPT come strumento ausiliario nei case studies ci rende più coinvolti e attenti."

Awareness and Responsibility in the Use of AI: per molti studenti, confrontarsi con la qualità delle risposte è stato un piccolo trauma che ha generato una maggiore consapevolezza sui limiti di un approccio superficiale. Una studentessa riporta: "all'inizio, utilizzare ChatGPT era tutt'altro che intuitivo: lo approcciavo come un "oracolo" da cui aspettarsi risposte definitive, forse anche un po' come un motore di ricerca. Tuttavia ho imparato a sfruttarlo come un vero e proprio compagno di studio, capace di stimolare il ragionamento critico e aiutarmi a colmare lacune che altrimenti sarebbero rimaste nascoste fino al momento dell'esame."

Interactive and Critical Learning: una studentessa ha rilevato un alto valore educativo nella costruzione di prompt sotto supervisione, apprezzando l'enfasi sull'uso corretto dell'IA: "Il messaggio principale che il professore ci ha trasmesso è che tutti oggi usano l'Intelligenza Artificiale, ma la differenza la fanno coloro che la usano nel modo giusto." Relativamente alla modalità di utilizzo una studentessa ha voluto rimarcare come i diversi metodi "didattici" abbiano avuto ciascuno una loro valenza specifica: "Un momento significativo è stato quando, durante l'ultima lezione, siamo stati invitati a "insegnare" all'IA, simulando il ruolo di educatori. Questo esercizio mi ha fatto capire l'importanza non solo di trasferire informazioni corrette, ma anche di farlo in un modo comprensibile ed efficace. In sintesi, l'integrazione di ChatGPT nel corso ha rappresentato un'esperienza formativa innovativa e stimolante, che mi ha aiutato non solo a padroneggiare gli argomenti del programma, ma anche a sviluppare competenze utili e trasferibili ad altri contesti, accademici e professionali."

Objective evaluation of results obtained: uno studente ha prodotto un'analisi dettagliata dell'esperimento, osservando come l'utilizzo del metodo "AI come Tutor" abbia portato ai migliori risultati. Ha notato: "Il gruppo che ha combinato pensiero autonomo con assistenza dell'IA, ha ottenuto risultati migliori rispetto agli altri gruppi. Senz'altro credo che il tool 'AI as a tutor' abbia dato dei benefici nel processo di apprendimento" e ha evidenziato come questa sperimentazione gli abbia dato nuovi stimoli per migliorare il suo processo di studio: "Per prepararmi all'esame conto di utilizzare l'AI come abbiamo fatto in classe, in particolare chiederò feedback e correzioni sulle mie risposte alle domande di recap a fine capitolo."

10.4 Conclusion, for now

L'integrazione dell'intelligenza artificiale generativa nella didattica universitaria si pone come una frontiera innovativa, capace di trasformare profondamente sia i processi di insegnamento che quelli di apprendimento. La sperimentazione condotta ha evidenziato come un approccio strutturato all'uso dell'IA, supportato da un design didattico mirato e consapevole, possa non solo facilitare l'acquisizione delle conoscenze tecniche, ma anche sviluppare competenze critiche e metodologiche in grado di arricchire il bagaglio intellettuale degli studenti. Il feedback degli studenti conferma il valore educativo di un'interazione guidata con l'IA, che ha incentivato lo sviluppo di abilità chiave come il prompt engineering, la capacità di valutazione critica e la consapevolezza dei limiti intrinseci delle tecnologie generative. In particolare, l'uso dell'IA nelle modalità di "tutor" e "consulente" ha fornito stimoli significativi alla didattica ed in generale al processo di apprendimento. La sperimentazione, pur limitata nella sua estensione, offre elementi che suggeriscono come un'integrazione sistematica e consapevole dell'IA possa contribuire a un'evoluzione sostanziale della didattica universitaria, in grado di adattarsi alle esigenze di un contesto accademico e professionale sempre più digitalizzato. La possibilità di personalizzare la base di conoscenze del bot con i contenuti specifici del corso abilita modalità di interazione e di didattica aumentata che sarebbero semplicemente impossibili in modalità tradizionale (a parità di efficienza). Pertanto ritengo che questa, unitamente allo sviluppo di agenti per aiutare nello sviluppo stesso di prompt "utili" ad avvicinare all'utilizzo sempre maggiormente consapevole, sia una frontiera da esplorare e da percorrere con fiducia.

11 Conclusions and future perspectives

Sono passati solo pochi mesi dal momento in cui ho iniziato una seria riflessione sull'argomento ma sono stati sufficienti per convincermi che l'Intelligenza Artificiale Generativa è destinata a diventare parte integrante del futuro dell'istruzione superiore. Siamo ancora nelle fasi iniziali (soprattutto in Italia) di questa trasformazione, ma è già possibile intravedere aule più interattive e coinvolgenti, in cui le lezioni frontali lasciano spazio a mentoring personalizzato, discussioni e progetti creativi, mentre l'IA gestisce alcune attività di routine in background. L'integrazione dell'IA nell'istruzione superiore non è semplicemente una questione di adozione tecnologica, ma di trasformazione identitaria per studenti e docenti. Gli studenti devono sviluppare non solo competenze tecniche nell'uso dell'IA, ma anche un'identità professionale che integri l'IA come strumento mentre valorizza il proprio contributo umano unico. Il 'gioco intenzionale' con queste tecnologie rappresenta un approccio particolarmente promettente: creando un ambiente sicuro in cui gli studenti possono sperimentare con l'IA, fallire, adattarsi e imparare, le istituzioni di istruzione superiore possono preparare i laureati non solo ad adattarsi alle tecnologie attuali, ma a prosperare in un ecosistema tecnologico in continua evoluzione. Questa preparazione olistica che abbraccia competenze tecniche, adattabilità, pensiero critico e consapevolezza delle proprie capacità unicamente umane rappresenta forse il contributo più significativo che l'istruzione superiore può offrire nell'era dell'intelligenza artificiale. If we keep student empowerment, inclusion, and educational ethics as our guiding star, we can adopt AI in ways that genuinely enrich university education without distorting it. History shows that, with past innovations (from printing to computers), educational systems can adapt and often emerge stronger from technological revolutions, provided they remain anchored to core principles of equity, integrity, and student centrality (regardless of, and in resistance to, the different agendas that politics, certain politics, would like to impose). What I have observed in my small Italian context is, in reality, part of a global phenomenon of impressive scale. A recent bibliometric analysis of more than 1,278 scientific publications on AI applications in higher education reveals the planetary scope of this transformation [11]. The numbers are striking: from just 18 publications in 2007 to 255 in 2022, with growth showing no sign of slowing down. What I find particularly meaningful is the geographical distribution of this interest: China leads with 31% of publications (468 papers), followed by Malaysia (12%, 189 publications). This is not accidental; it reflects strategic national investments in integrating AI into education that go far beyond the individual experiments of single instructors or institutions. We are witnessing a global race to redefine higher education, and Italy, unfortunately, does not seem to be among the protagonists. International research has identified AI applications ranging from automating administrative teaching tasks (freeing faculty time for more creative activities) to personalized student support through intelligent tutoring, from augmented reality for immersive experiences to data-driven approaches for adaptive education. What emerges is a technological-educational ecosystem far richer and more articulated than our local experiments might suggest. There is, however, an even more significant point: this exponential growth in research reflects the recognition that integrating AI into education is a multidisciplinary and interdisciplinary field. It is no longer about "adding technology" to existing educational practices, but about fundamentally rethinking what it means to teach and learn in the age of artificial intelligence. Recommendations across the international literature converge on one point: we need a systemic change in teaching methods through computerization and digitization that leads to the development of a true "digital pedagogy". This global context forces me to reflect on the responsibility we have as Italian educators. We cannot afford to remain spectators of a transformation that is redefining higher education worldwide. My small experiments with Centaur and Cyborg approaches, as limited as they are, fit within a broader movement of pedagogical rethinking. But it is clear that we need a change of scale: from individual experimentation to institutional policies, from sporadic initiatives to systematic investments in faculty training and in the development of appropriate technological infrastructures. The good news is that we have an enormous body of international research to draw from, others' mistakes to avoid, and good practices to adapt to our context. Delay can become an advantage if we are able to learn from global experiences and develop approaches that value the specific strengths of our educational tradition while integrating the potential of AI. This integration process is hard work and requires a trial-and-error approach, but early field experiences suggest that the potential benefits are worth the investment. The transformation of higher education through AI is not simply a technological upgrade, but a deep redefinition of the educational process itself and of the educator's role, which becomes more dynamic, multifaceted, and potentially more meaningful than ever.

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